Cantilever: HOLLOW DESIGN ACCEPTED

```
cshell design accepted: mass saving 2.4% (V/V0 0.9760); K/K0 1.0200 / 1.0196, P/P0 1.0336 / 1.0337
(optimisation / shifted grid) ≥ 1, export mass-matched and watertight

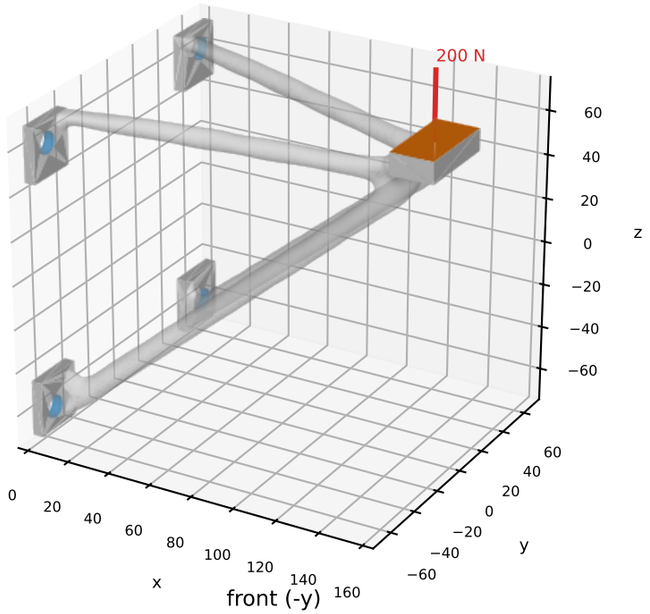
mass      target 61.86 g   final 61.85 g   error -9.9165e-06
optimiser grid  mass saving 0.02403 (V/V0 0.976)  K/K0 1.02  P/P0 1.034  governing buckling  feasible True   (accept: K/K0 >=
    1, P/P0 >= 1)
shifted grid    mass saving 0.02403 (V/V0 0.976)  K/K0 1.02  P/P0 1.034  governing buckling  feasible True   (accept: K/K0 >=
    1, P/P0 >= 1)

export    h 0.2 mm  faces 2387104  watertight True  volume 5.948e+04 mm3  material bodies 1  shells 7  enclosed voids 6
    escape holes 0
wall [mm] min 1.435  p05 1.571  median 5  max 41.51
closed_area_frac   0.513
hollow_area_frac   0.324

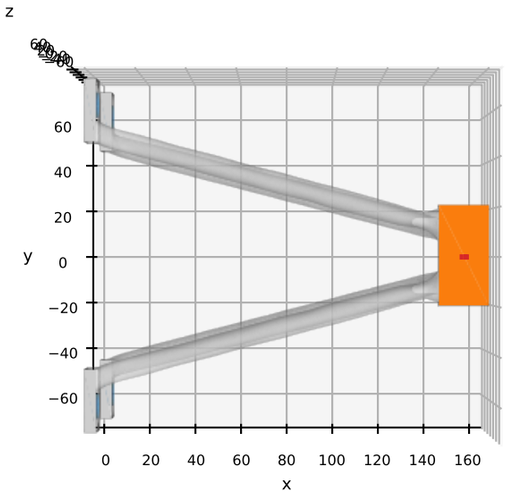
optimiser: {'method': 'mma', 'starts': ['solid'], 'max_iter': [60, 80, 40], 'move': 0.1, 'tol_x': 0.005, 'tol_f': 0.0001,
    'mass_tol': 0.0001, 'sliver_weight': [0.0, [0.0, 0.05], 0.1], 'seed': 0}
grid: {'phases': [{'h_e': 0.8, 'h_g': 0.2}, {'h_e': 0.4, 'h_g': 0.1}], 'export_h_mm': 0.05, 'rho_min': 1e-06,
    'tessellation_mm': 0.02, 'active_margin_mm': 0.5, 'geometry_device': 'auto', 'geometry_offload': True}
growth: {'enabled': True, 'phase': 0, 'tol': 0.001, 'patience': None, 'quantiles': [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8,
    0.9, 1.0], 'step_tol': 0.0001, 'max_iters_per_step': 25, 'min_iters_per_step': 2, 'samples': 16}
objective: {'type': 'lightweight', 'lw_margin_stiffness': 1.02, 'lw_margin_capacity': 1.02, 'constraints': [],
    'constraint_scale': 0.01, 'feasibility_tol': 0.01, 'w_stiffness': 0.5, 'w_capacity': 0.5, 'min_gain': 0.02,
    'floor_stiffness': 0.85, 'floor_capacity': 0.85, 'floor_strength': None, 'floor_buckling': None, 'floors_as_constraints':
    True}
export cfg: {'h_mm': 0.2, 'device': 'gpu', 'closed_thr': 0.9, 'escape_holes': False}
```

Cantilever: supports (blue, all DOF fixed; 4 patches) and loads (orange faces; 1 red arrow, one per loaded patch at its area centroid, length and label = its share of the total 200 N). Faces read from bc_faces.npz (as analysed).

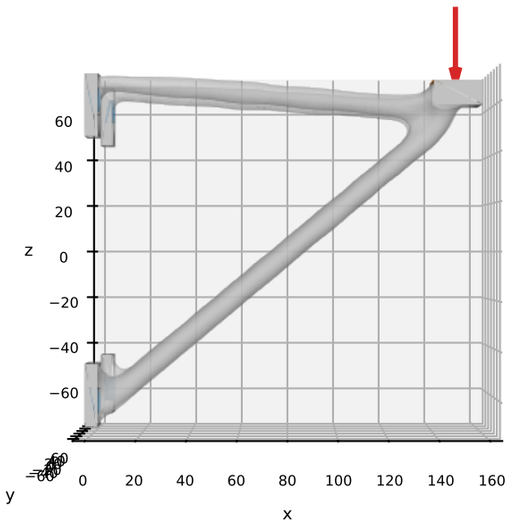
isometric



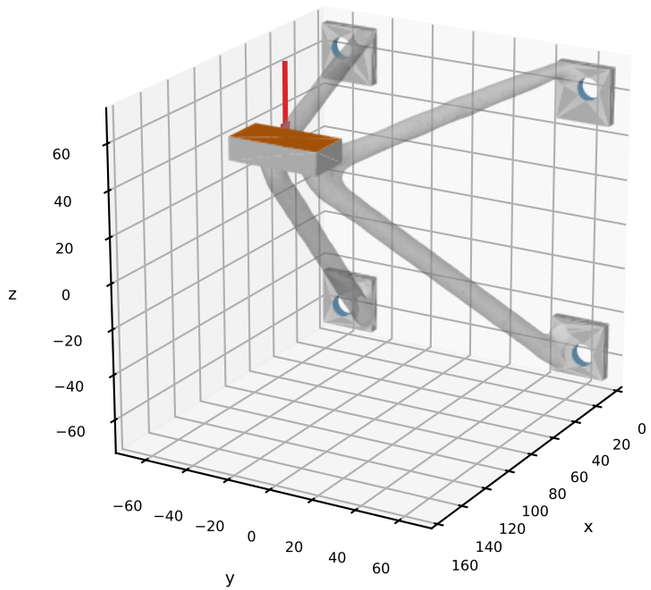
top (+z)



front (-y)

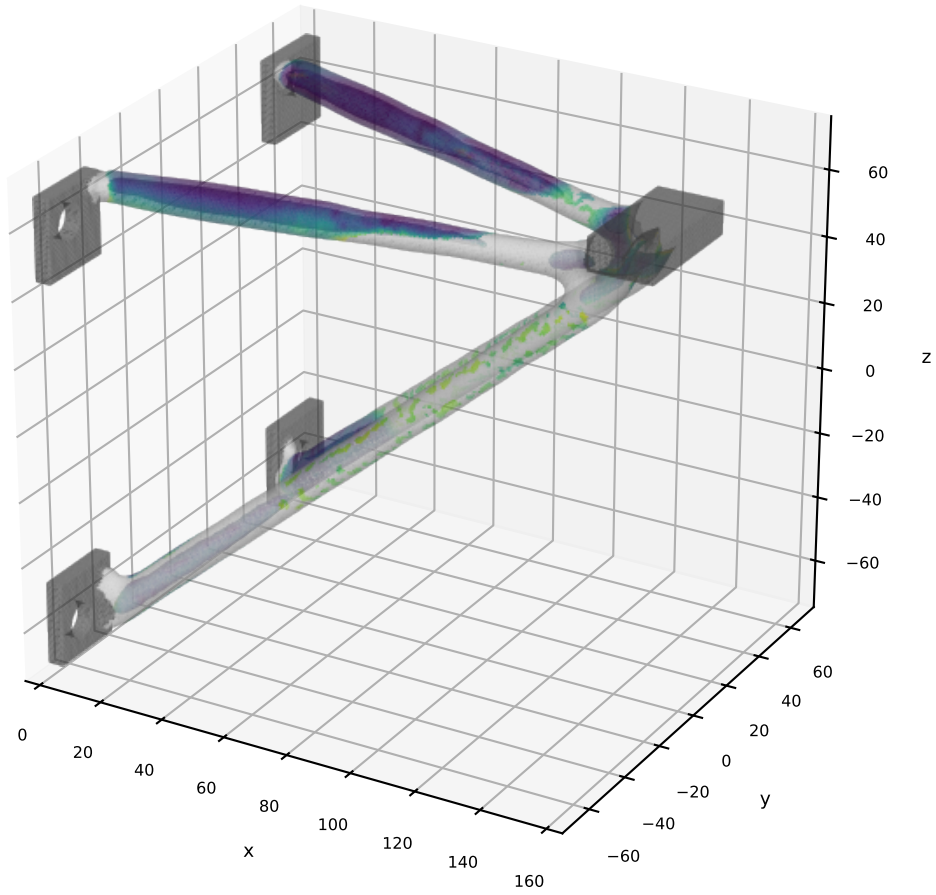


rear isometric

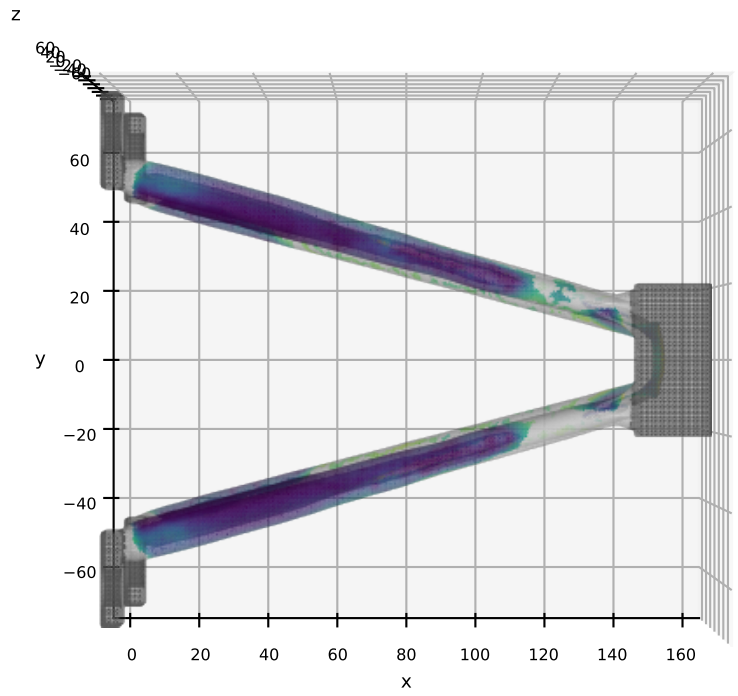


Cantilever: final design (STL 2,387,104 faces, shown decimated to 164,274; wall thickness from fields.vtp)

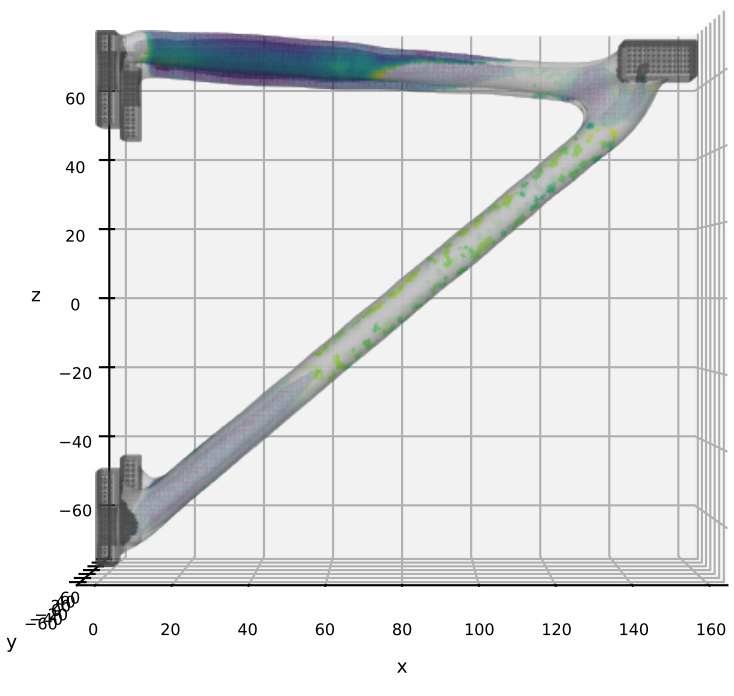
isometric



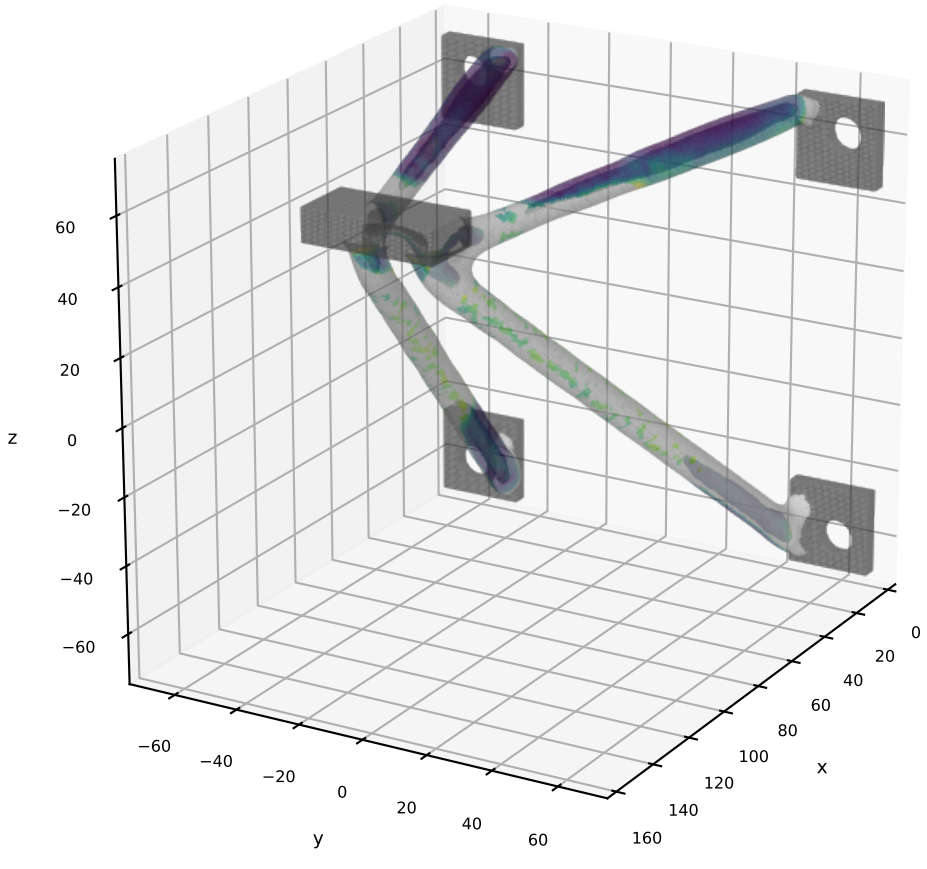
top (+z)

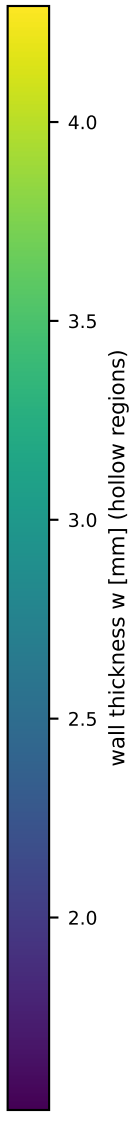


front (-y)



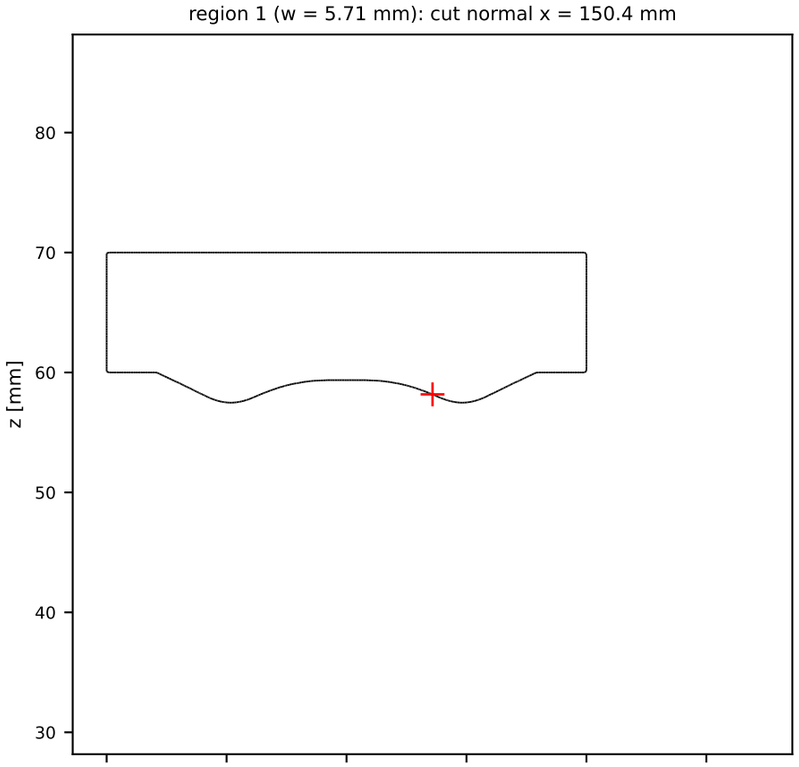
rear isometric

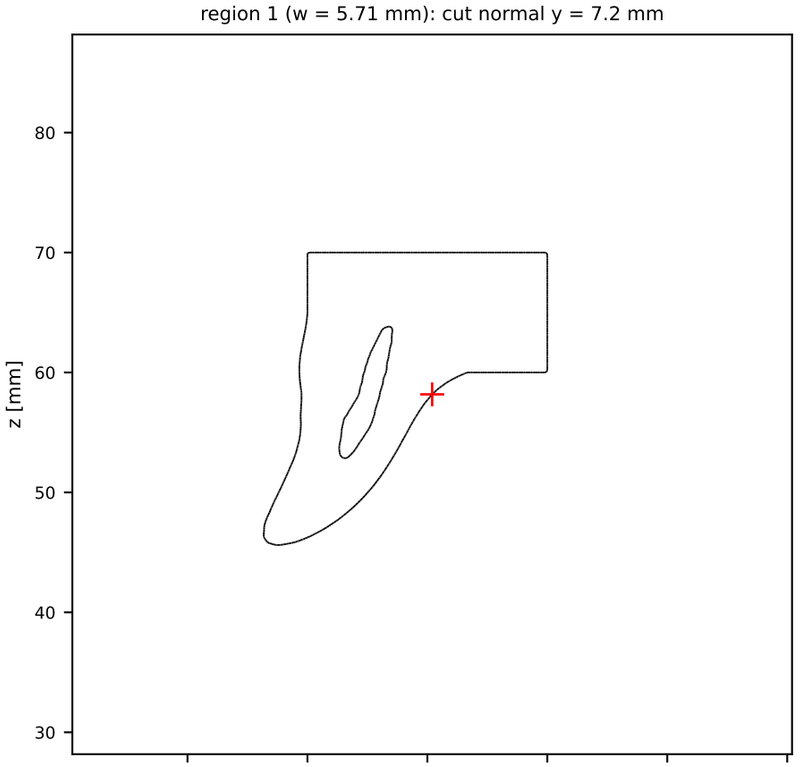


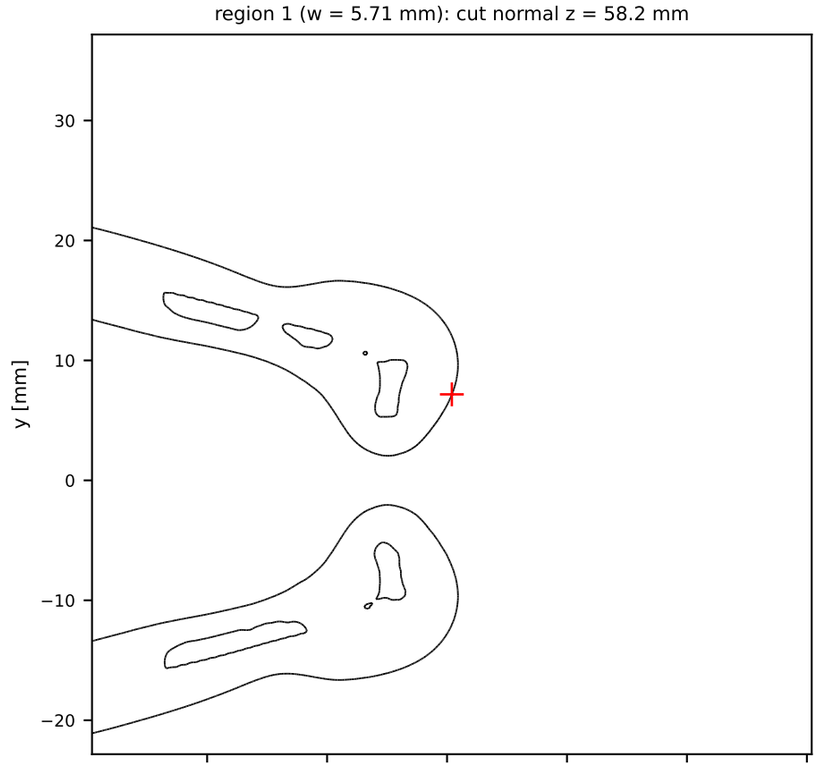


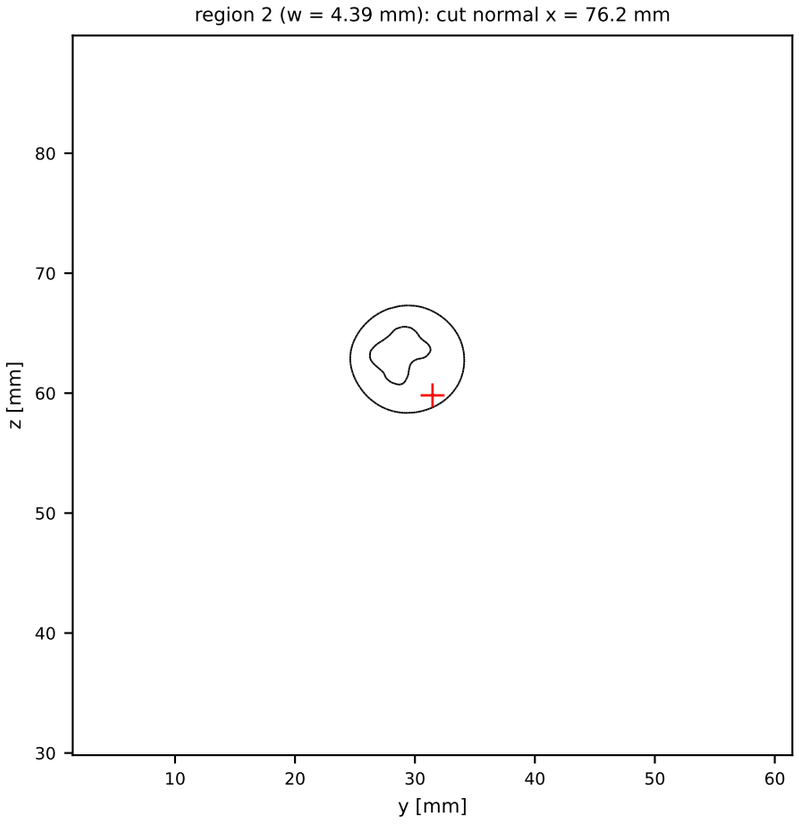
light grey = solid (closed) regions; dark grey = mounts; the colourbar covers the hollow regions only

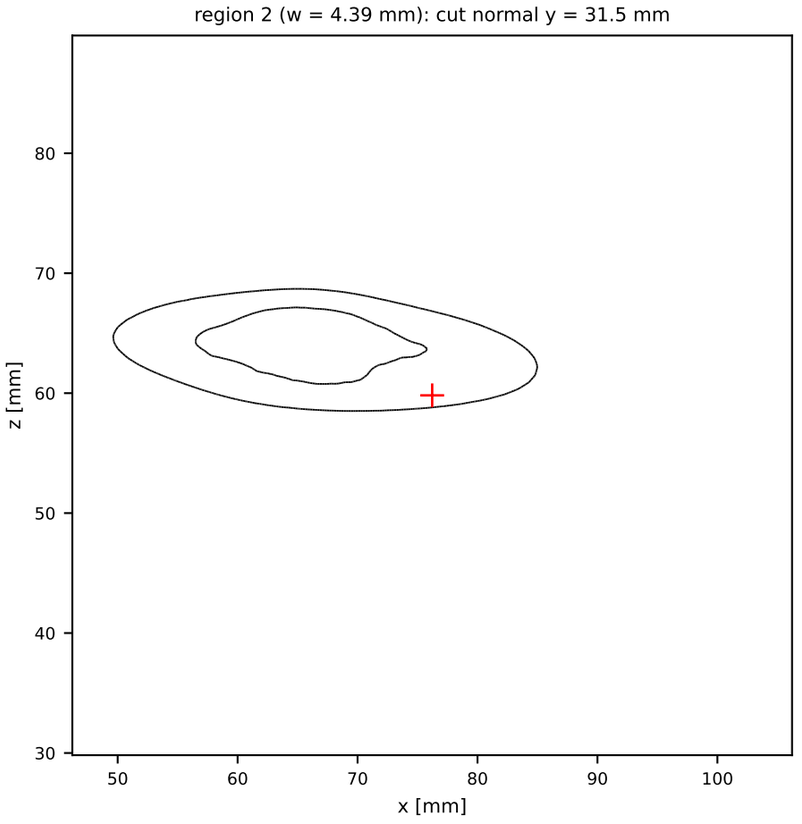
Cantilever: sections of the final design through the two thickest-wall regions (red +); +-30 mm window

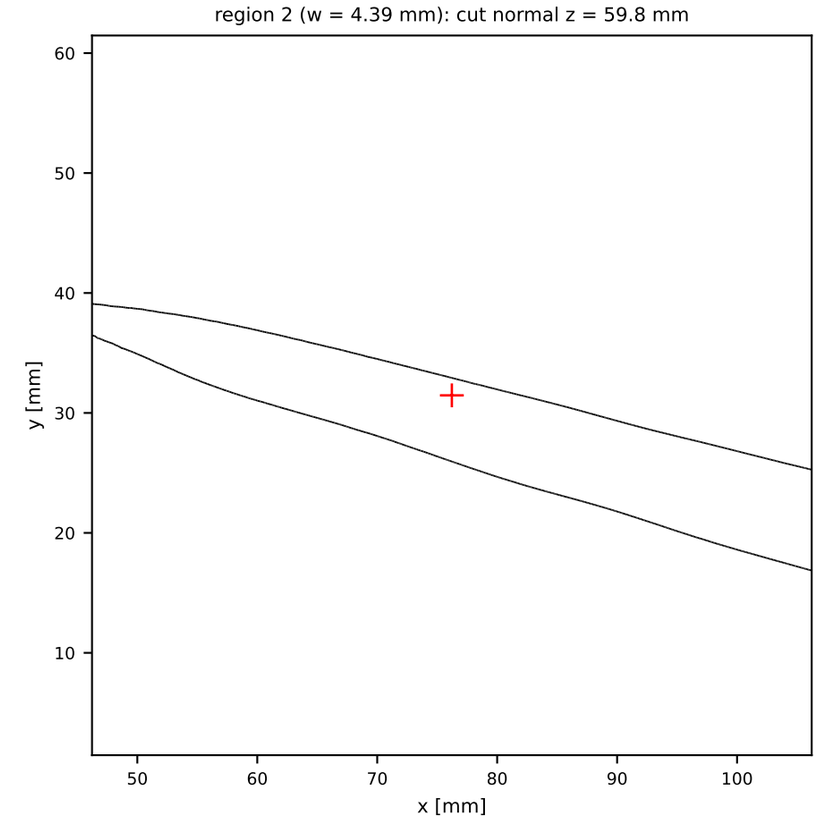






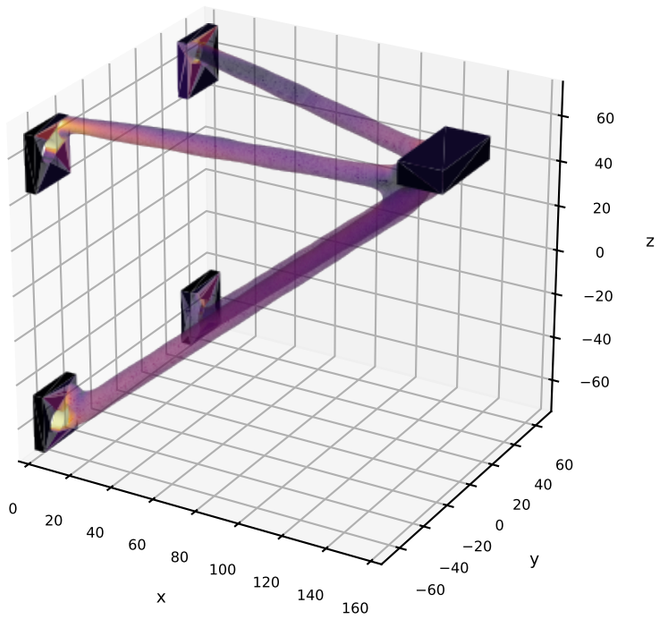




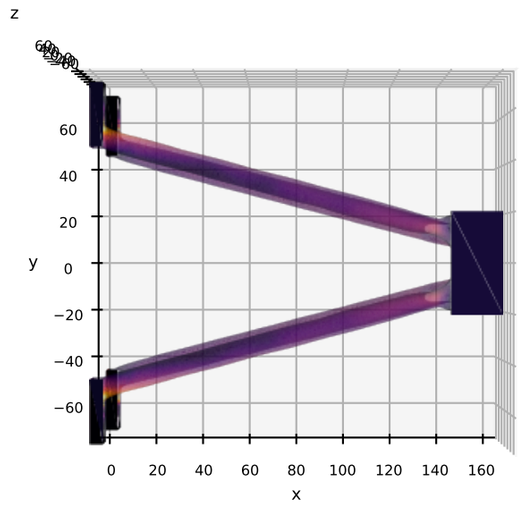


Cantilever: von Mises stress, original solid vs final design (peak solid 20.2, final 16.8 MPa; values from fields.vtp, nearest point)

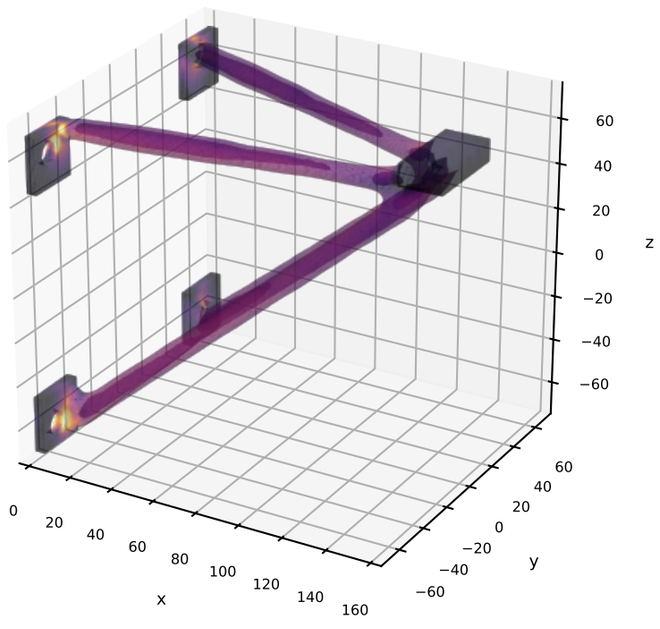
solid: isometric



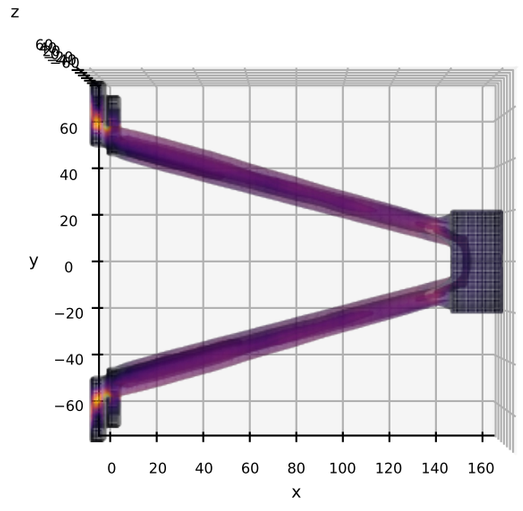
solid: top (+z)

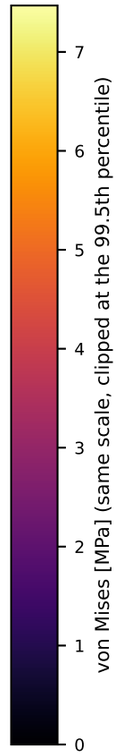


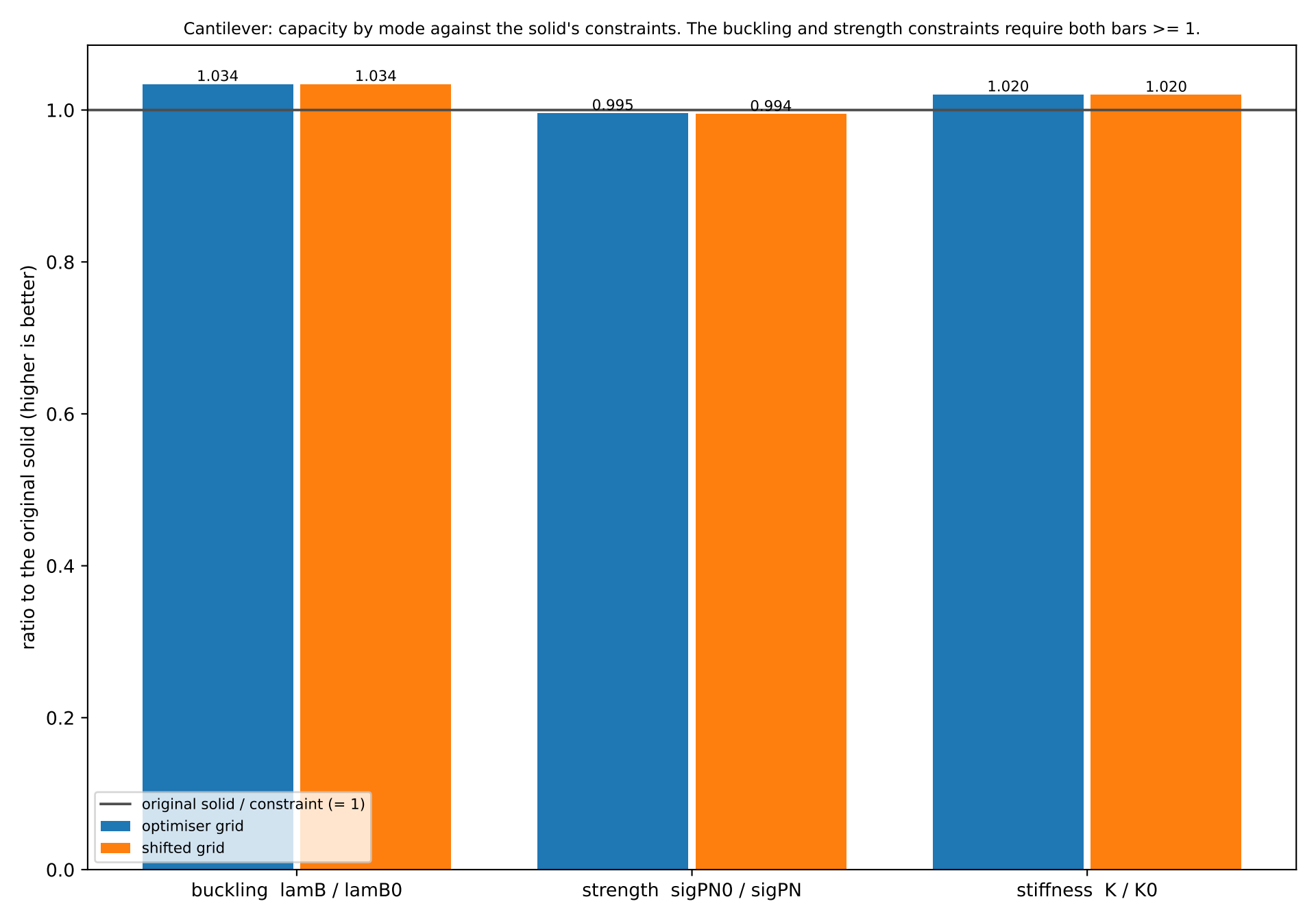
final design: isometric

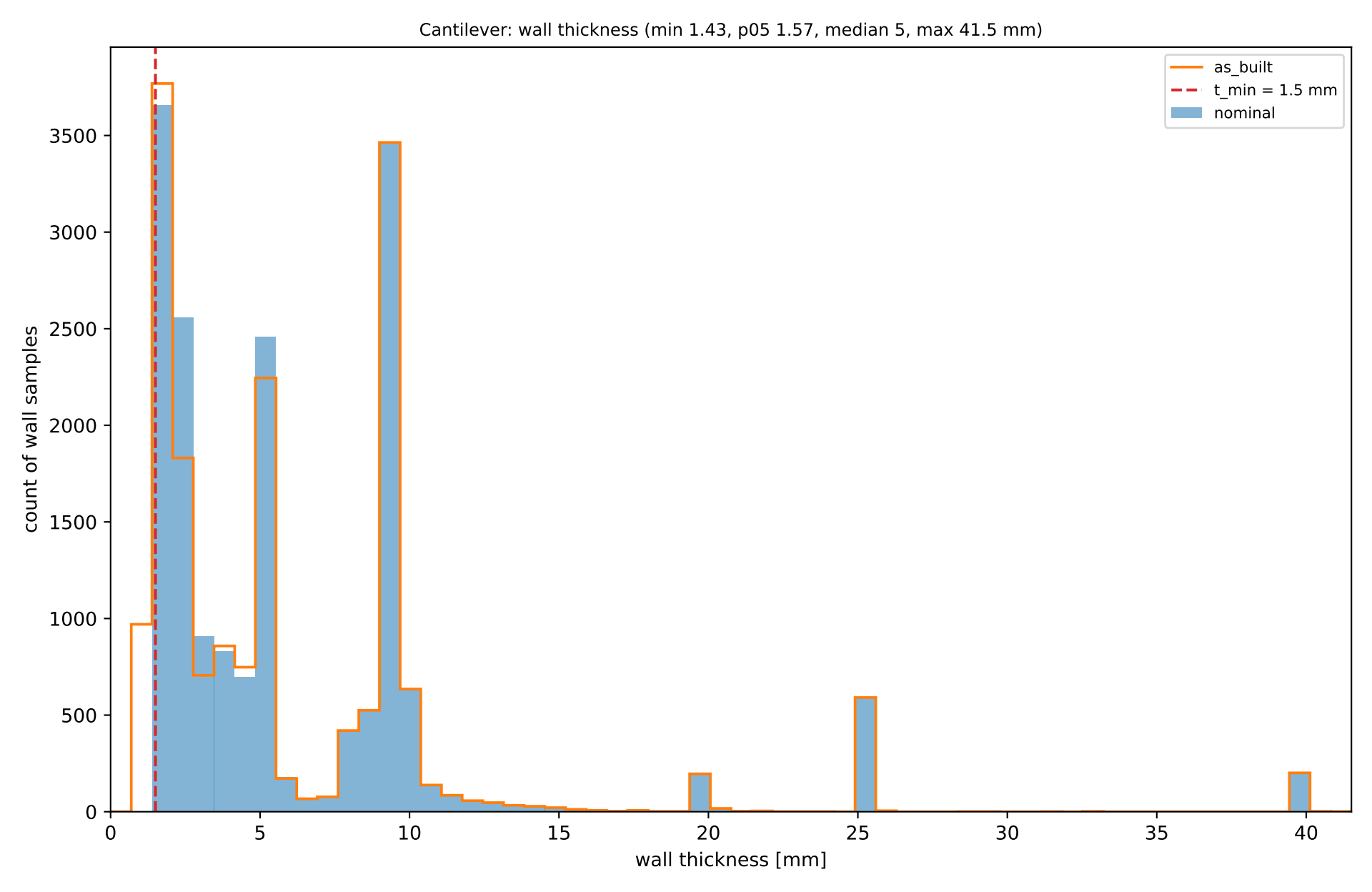


final design: top (+z)

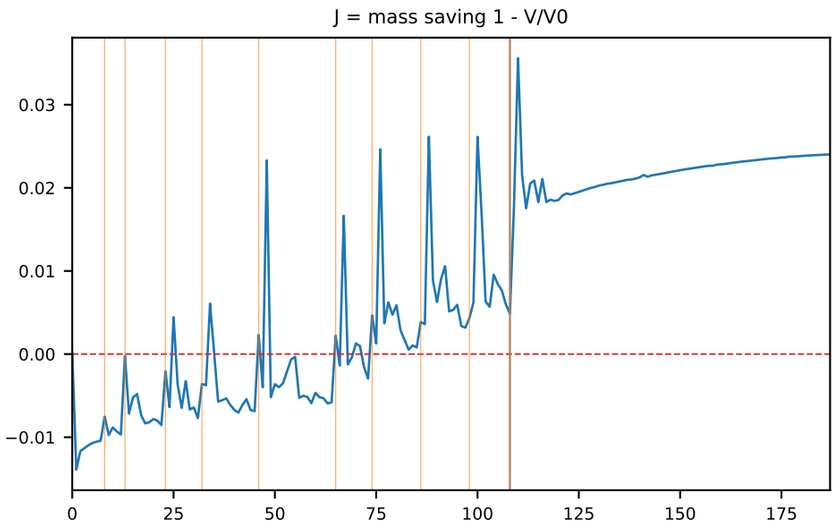


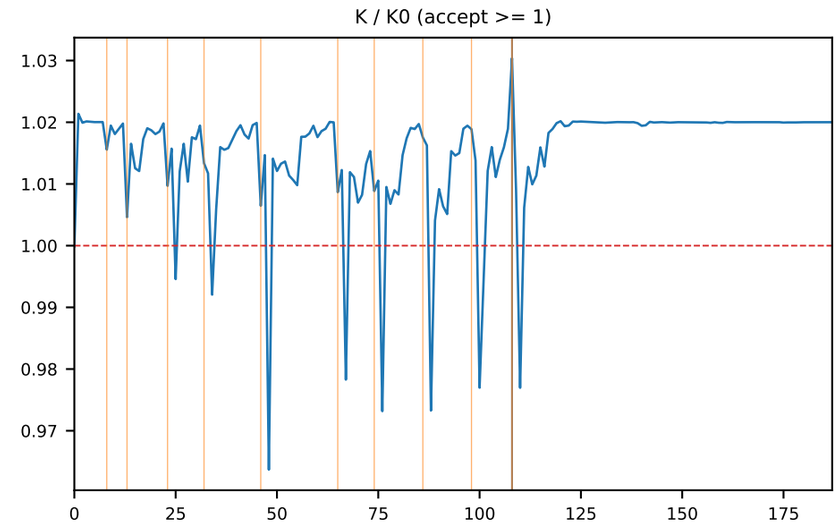


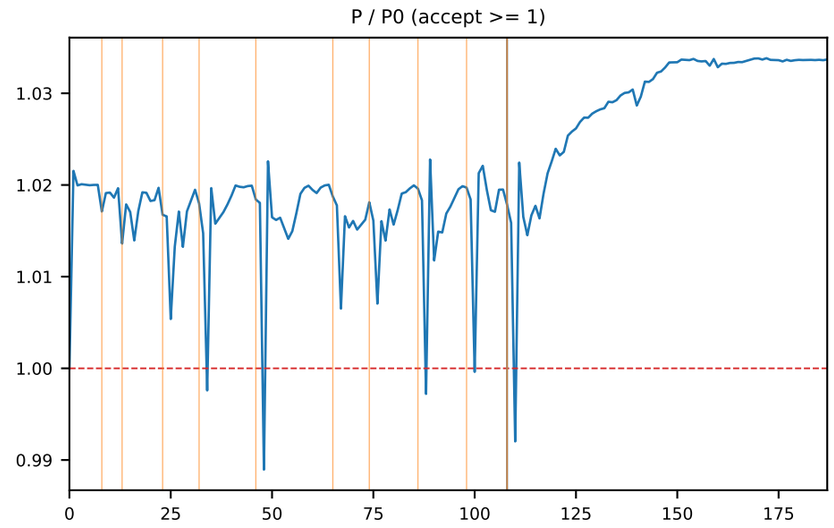

Cantilever: capacity by mode against the solid's constraints. The buckling and strength constraints require both bars >= 1.
ratio to the original solid (higher is better)
0.0
0.2
0.4
0.6
0.8
1.0
1.034
1.034
0.995
0.994
1.020
1.020
original solid / constraint (= 1)
optimiser grid
shifted grid
buckling lamB / lamB0
strength sigPN0 / sigPN
stiffness K / K0

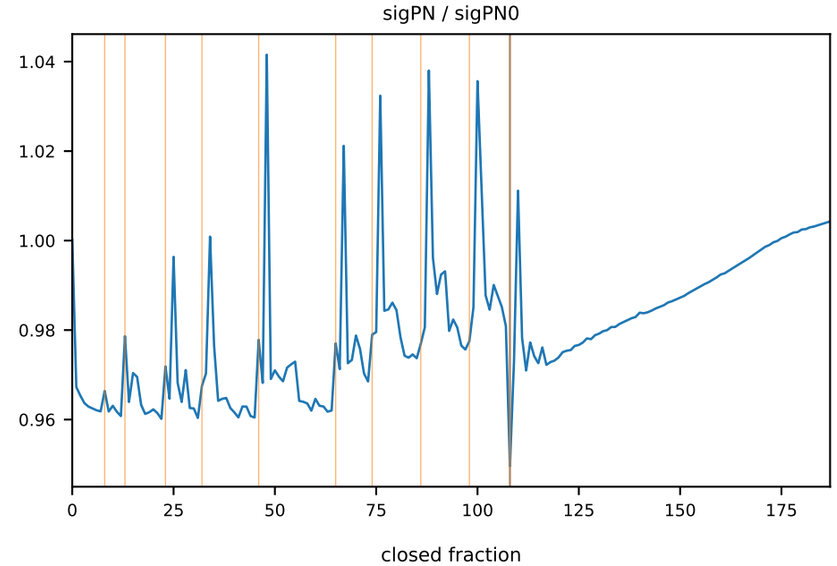

Cantilever: wall thickness (min 1.43, p05 1.57, median 5, max 41.5 mm)
as_built
t_min = 1.5 mm
nominal
count of wall samples
wall thickness [mm]
0
500
1000
1500
2000
2500
3000
3500
0
5
10
15
20
25
30
35
40

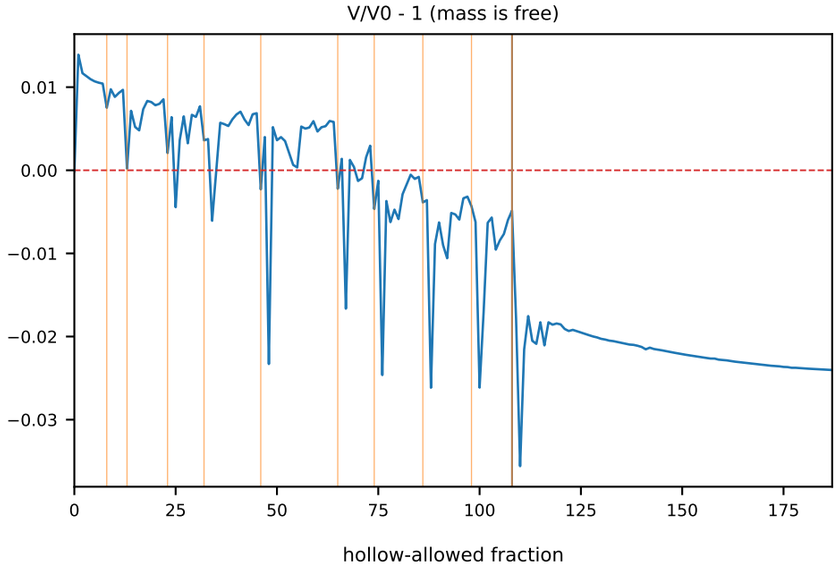
Cantilever: convergence history

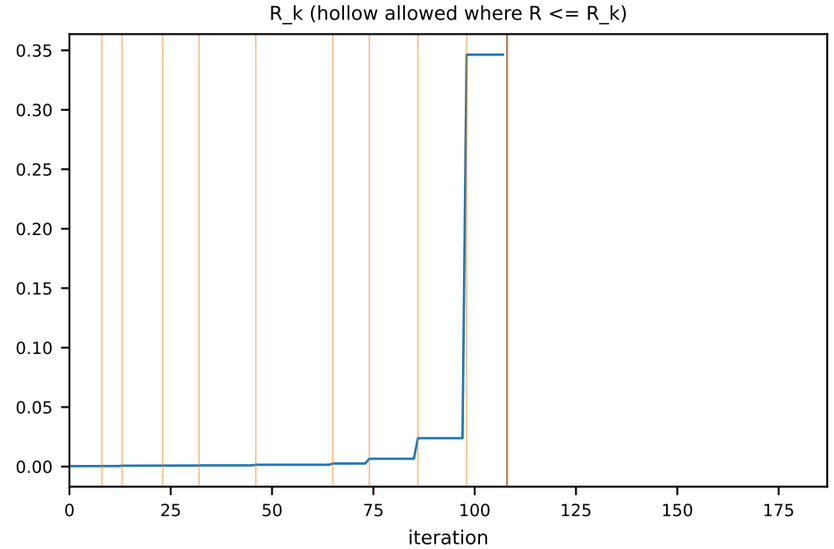


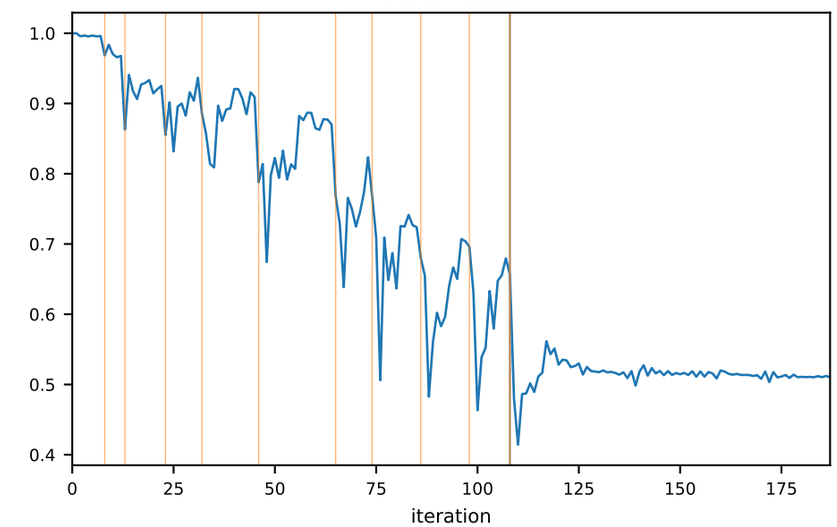


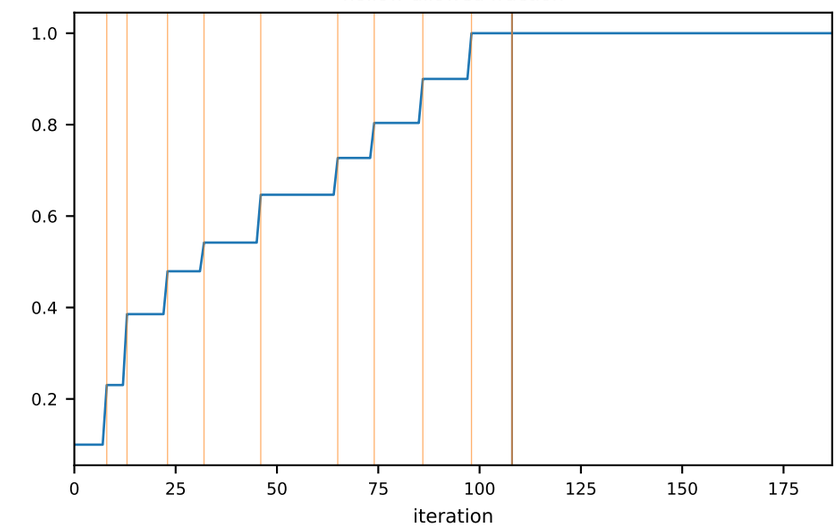








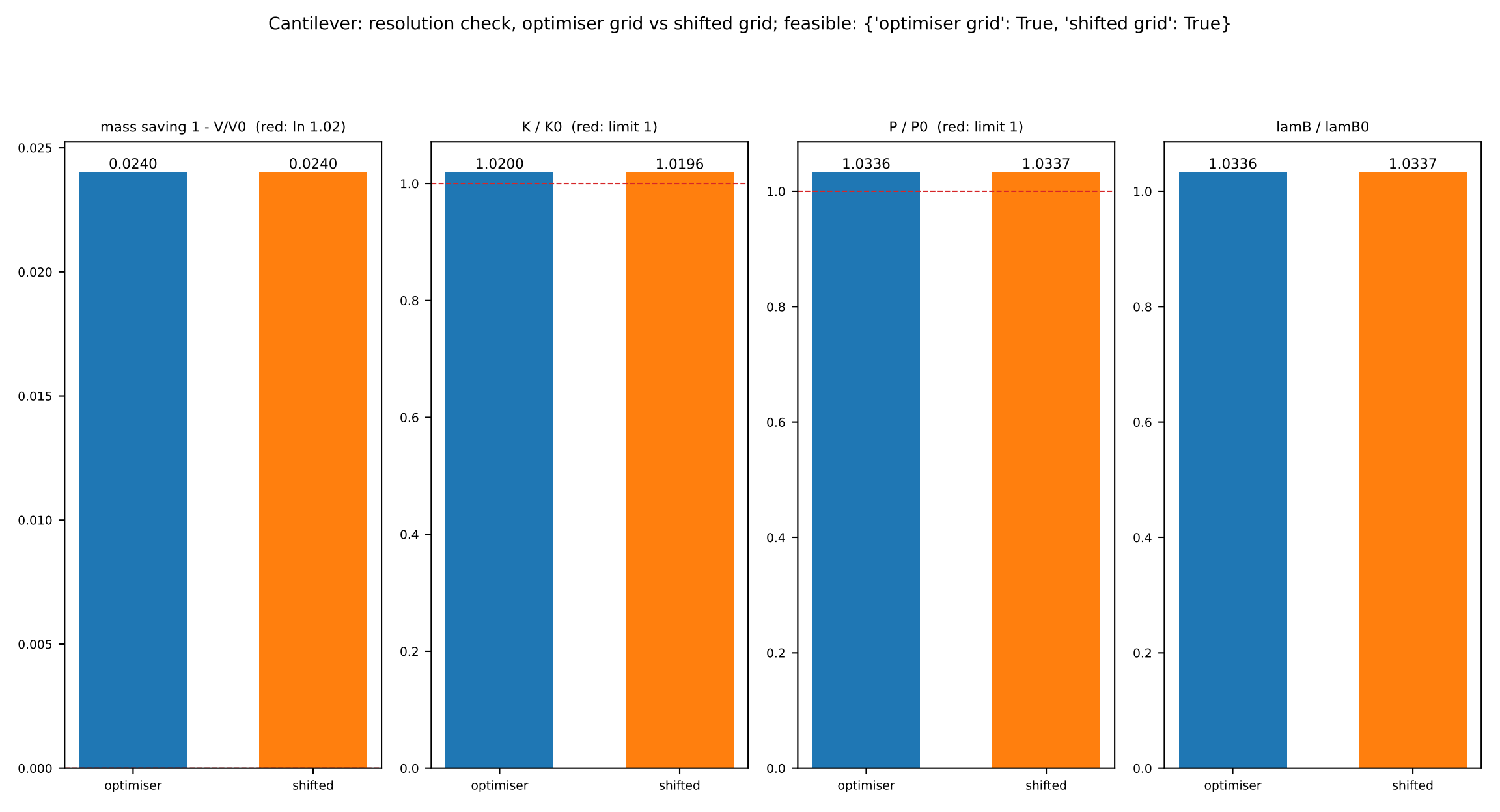


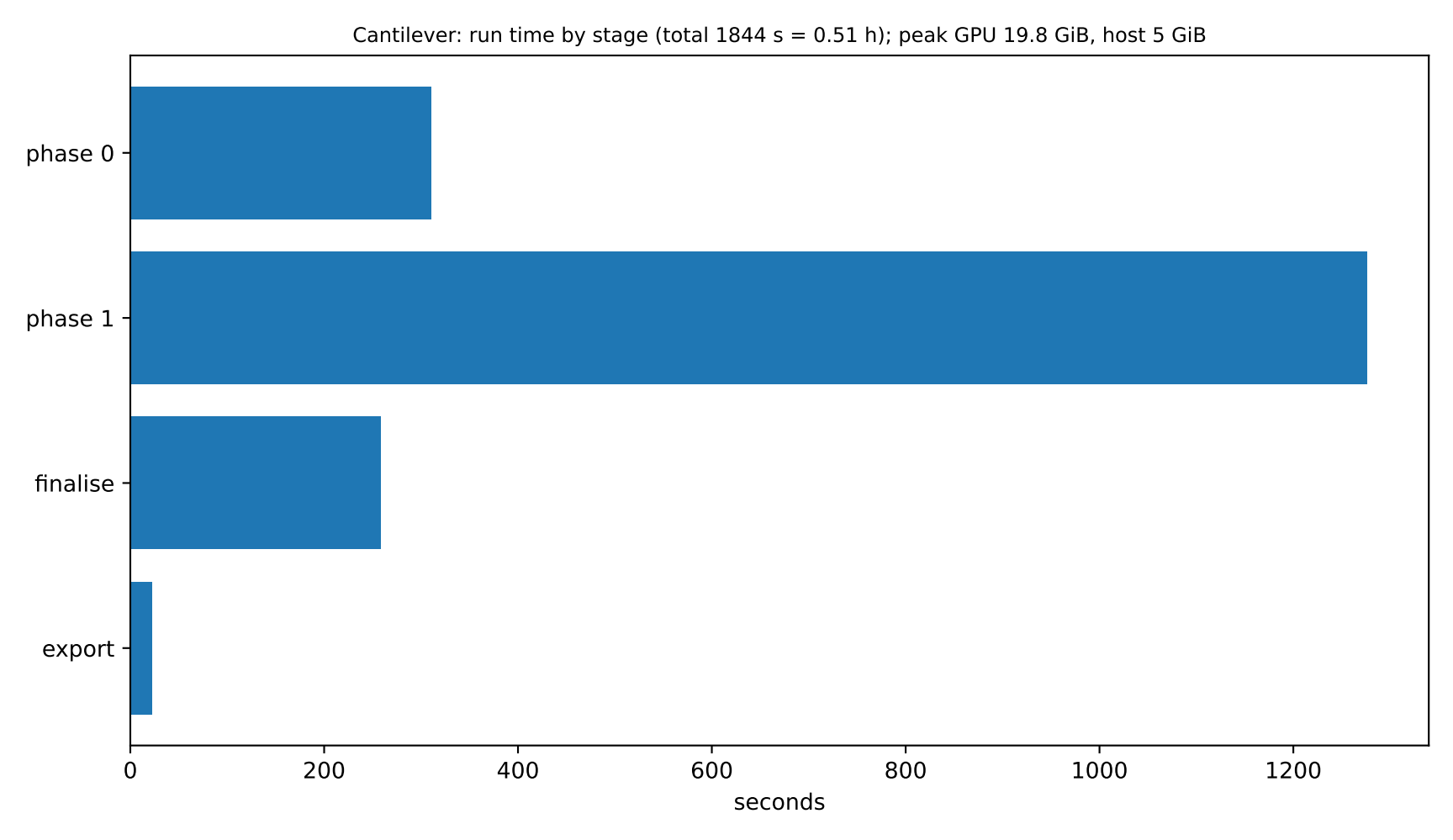


grey vertical: phase boundary
orange vertical: growth step (R_k rise)
iterations 188; growth steps 10
red dashed: constraint / acceptance level

```
enclosed voids: 6;  escape holes: 0;  voids without a hole: 6
  void 0: volume 736.3 mm3, holes 0
  void 1: volume 2229.6 mm3, holes 0
  void 2: volume 646.77 mm3, holes 0
  void 3: volume 2214.8 mm3, holes 0
  void 4: volume 554.04 mm3, holes 0
  void 5: volume 87.92 mm3, holes 0

Escape holes sit at the lowest-stress wall points of each void (outward direction in escape_holes.csv).
```


Cantilever: resolution check, optimiser grid vs shifted grid; feasible: {'optimiser grid': True, 'shifted grid': True}
mass saving 1 - V/V0 (red: ln 1.02)
0.0240
0.0240
K / K0 (red: limit 1)
1.0200
1.0196
P / P0 (red: limit 1)
1.0336
1.0337
lamB / lamB0
1.0336
1.0337
0.025
0.020
0.015
0.010
0.005
0.000
1.0
0.8
0.6
0.4
0.2
0.0
optimiser
shifted


Cantilever: run time by stage (total 1844 s = 0.51 h); peak GPU 19.8 GiB, host 5 GiB
phase 0
phase 1
finalise
export
0
200
400
600
800
1000
1200
seconds